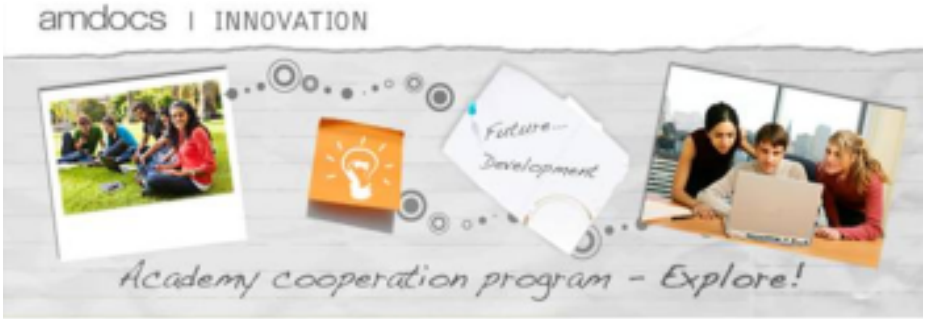
CAT · Research · GSS · innovation · Explore · Academia

The case of GSS - analyzing customers’ capacity growth needs



With the new “Explore! Academia” program, Amdocs GSS benefited from cooperation with PhD students in Tel-Aviv University, in the area of Capacity Analysis of Trends (CAT). CAT improved the forecasting of production hardware capacity in two large GSS projects in APAC, contributing to $10 Million savings in the first implementation of this research.

Just a few of the many benefits of the Explore! Academia program:

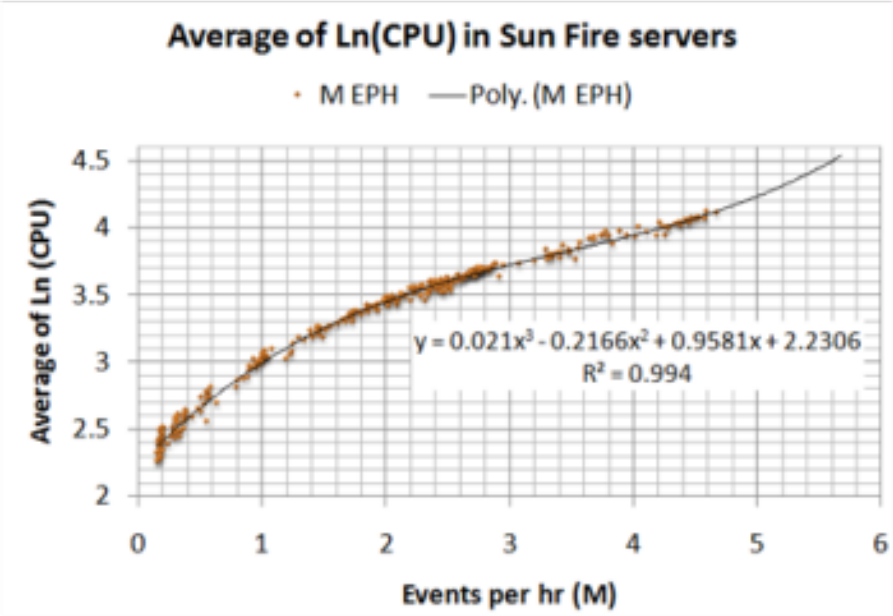
- Introducing the Amdocs brand in universities
- Encouraging research in domains which are strategically important to Amdocs
- Utilizing academic achievements to build cutting edge technologies
- Ensuring that Amdocs stays at the forefront of the industry
- Improving business prospects for projects created within Amdocs.

GSS-PSO Scientific Officer Gilli Shama, under the supervision of Zeev Berkowitz, VP GSS Operations (GSO) PSO, recognized this opportunity “Explore! Academia” offers, and suggested implementing an operations research approach, using

advanced statistical methods, for analyzing capacity growth needs for GSS customers.

Sasha Alexander Apartsin, a Computer Science PhD student from Tel-Aviv University brought with him a professional research lead with cutting edge statistical analysis approach, combined with his previous work experience in telecom infra. The research project was done in collaboration with GSSO Infra Capacity team, PBG HW Lab, Delivery TACT team and GSSO APAC accounts.

The Capacity Analysis of Trends (CAT) model for Amdocs Online Charging (OLC) revealed an extremely tight connection between the events entering a Rater and Balance server, and the server's CPU utilization. The CPU utilization distribution was found to be log-normal, where the average CPU and the standard derivation are a function of the number of events. This finding led to realize that average CPU growth is exponential, rather than linear. Knowing the distribution, the server's success rate was predicted.



The success rate prediction raised the understanding and forecasting of hardware capacity limits:

- Knowing the limits, we can expect when and if penalties will be reached, and plan mitigation accordingly

- CAT offers well sustained arguments for our capacity capabilities, thus raising customer's perception of Amdocs as an innovative and professional company

The CAT model predicted the growth of required resources in OLC servers on two large GSS projects in APAC. In one managed services project, we used CAT as part of our analysis, which led to a significant $10M hardware saving. On another manged services project, CAT was used internally to compare SOAK Test results on a new site with the results from production site. GSSO Infra Capacity team is expanding the CAT approach to its next generation, as a practical tool for automatic capacity cost estimation in proposals. Gilli Shama defines the collaboration with the PhD students program as a great success, and the analysis technique of the student research will be used in Amdocs GSS' further capacity growth analyses.

About Explore!

"Explore!" is a new innovation program that establishes a comprehensive framework for Amdocs' collaboration with the academia worldwide. The program encompasses all forms of collaboration with the academia, including joint research projects, innovation labs, course student projects, internships, and more. Hanoch Sapoznikov leads the overall program. Many more details about the Explore! program can be found at: Innovation/Explore!